PromoFinder

A tool that reads a microdrama's screenplays and hands an editor a ranked list of 20-60 second promos, with exact cut points.

Live app: kuku-promo-finder.streamlit.app · Code: github.com/vatsal3512/kuku-promo-finder · Raw outputs: raw_outputs/ in the repo

Which problem, and why

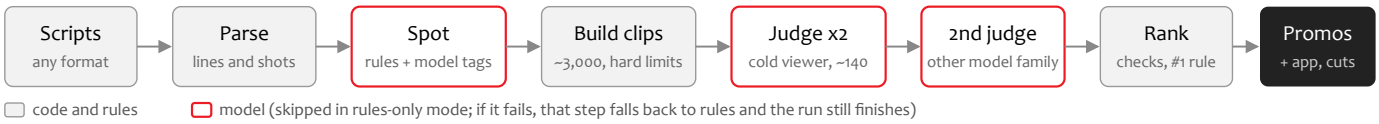
Problem 1. Today an editor watches every episode to find promo moments, about 100 minutes of footage for a 50-episode show, and Kuku releases new shows every week. The output is easy to check (a timestamp either works or it doesn't), the brief defines a good promo clearly, and promos are how paid viewers arrive, so better picks are worth real money. The editor stays in charge and still approves every cut, but starts from a ranked shortlist instead of raw footage.

What I assumed

- Script timestamps match the finished video, and CAMERA lines mark real shot changes, so cuts land on them.
- A promo's job is to bring in new viewers, so early-episode clips rank higher (viewers land near the start). Big late twists are kept but marked for retargeting existing viewers.
- "Hook in the first 3 seconds" means the hook starts within 3 s of the cut. "Ends on a cliffhanger" means the cut lands within 1.5 s after that line, never mid-line.
- "Makes sense on its own" is judged by a model that sees only the clip, never the show. Ad risks (violence, gore, coercion, minors) are flagged for a person, not silently removed.
- Free tiers only, and every model is one config value, so Kuku's own models can be plugged in.

How it works

The model never writes a timestamp. It points at lines in the script; code works out the cuts. I tested this: when the same model wrote timestamps itself, 23 of 100 promos were broken, and 22 of those stopped at the *start* of the cliffhanger line, cutting off the very moment the promo exists for.



- **Hard limits in code:** 20-60 s, hook within 3 s, cut right after the cliffhanger, never mid-line. A clip under 3/5 on "makes sense alone" or on cliffhanger is dropped.
- **Craft checks the judges miss** (they read the whole clip at once): no spoken line in the first 3 s, a one-word opening, a long silent tail, length over 50 s.
- **Ranking:** one promo per scene, pairwise model comparisons for the order. #1, the one to post first, must come from the first 20% of episodes, have a strong hook and cliffhanger, and carry the fewest ad flags.
- **Output per promo:** segments, hook, cliffhanger, why it works, why it has that rank, the episode it sends viewers to, ad flags; plus an edit list and ffmpeg cut scripts. The app shows each promo with a timeline and transcript, and the editor approves or rejects it.

> **Tech stack:** Python 3.12 · Pydantic (strict but forgiving reply schemas) · OpenAI-compatible client, any provider by one string · NVIDIA NIM free tier: Nemotron 3 Ultra (picks, judges), Mistral Large 4 (second judge), MiniMax M3.1 (held-out auditor) · Streamlit app on Streamlit Community Cloud · pytest (41 tests, no key needed) · GitHub.

How it scales, and what it costs

A 50-episode show takes about 3 minutes in **fast** mode (one judging pass) and 20-25 minutes in **full** mode (two passes, second judge, ordering), on NVIDIA's free tier. Full mode is about 140 model calls and 0.45M tokens; at paid rates of $1-3 per million tokens that is under $1.50 a show. Rules-only mode takes 5 seconds and costs nothing. Only tagging grows with episode count; judging is a fixed budget per show. Every answer is cached, so a crashed or changed run only pays for what's new. Shows run side by side; the limit is provider rate limits, so the tool rotates API keys, backs off when throttled, and stops at once on a bad key or empty quota. The editor reads about 12 minutes of shortlisted clips instead of 100 minutes of footage, and their approvals are logged to tune weights later.

How I know it's good

Check	What I found
Are the cuts usable?	15 of 15 in every run. The model writing timestamps itself: 77 of 100.
Do outside judges agree?	Two models that never picked anything scored the lists blind (MiniMax M3.1 took no part in any run). Against the rules-only list, the final list had better hooks (4.53 vs 4.33) and cliffhangers (4.67 vs 4.47), and every promo passed the standalone check (87% for rules only; Mistral: 100% vs 93%).

Did self-grading fool me?	Yes, once. A mid-day "improvement" looked better to the picking model (0.76 to 0.85) but worse to Mistral (standalone pass 100% to 80%). That's why the second judge exists; with it the list went back to MiniMax's best score (0.80).
What does a person think?	Blind A/B, 8 pairs, tool cut vs rules-only cut: I preferred the tool's in 5 of 8. One person, so a hint, not proof. My three "no" votes became the craft checks.
Will it hold up on a new show?	Tested on two made-up shows in other genres and formats (a whole series in one SRT-style file; Hindi speaker names with plain-seconds times): all promos valid. Broken files (empty, UTF-16, no times) and fake models that fail like real ones (bad key, no credit, refusals, odd replies) all finish with output. A later robustness release scores 0.78 vs 0.80 on the same blind check, inside judge noise, so the submitted list stays official.

The top of the final list

#	Clip	Opens on / cuts on	Why it's here
1	Ep1 00:39-01:30, 52 s	“Because he is marrying me.” Ends as her father refuses to look at her.	The whole premise: jilted at the altar for her own sister. Nothing for ad review, lands viewers on episode 1. Post first.
2	Ep5 00:54-01:45, 52 s	Her mother's book thrown into the rain. Ends on the stranger's “Come.”	Works with the sound off; the scarred stranger steps in for her the first time.
3	Ep12 00:26-01:09, 43 s	“You planted it!” Ends on “Fail, or refuse, and your husband hangs.”	The sharpest choice in the show. Needs an ad-policy check (bare hands in thorns).
4	Ep3 01:24-02:15, 52 s	A stranger floors the guard. Ends as he kneels with a ring in the church.	The only early clip with the soldier from the title; ends on “will she say yes?”.
5	Ep11 00:16-01:15, 58 s	“Your husband stole my gold pocket watch!” Ends mid-search of their home.	Easy for a stranger to follow, but long at 58 s; test it against the shorter ones.

Where it picks badly

It reads; viewers watch. All three judges gave openings like a bare "No." or a silent hand on a bedcover (first word at 0:09) four or five out of five, because they could see the tension coming later. The craft checks catch the measurable cases; whether a shot stops a thumb needs the video.

The judges are generous and noisy. Most scores are 4 or 5, and the same clip can move a point between batches. Once my changes moved scores by less than that, I stopped tuning against the judges.

It repeats storylines. One-per-scene stops repeated scenes, not storylines: four of fifteen promos are the poison plot. Three fixes I tried didn't separate storylines reliably.

Stage directions give things away. "She gasps, realizing her secret identity has been laid bare" tells the judge something a viewer never sees. The real fix for these is the video: check the first 3 seconds, then use hook-rate data once the ads run.

AI tools I used, and where they got it wrong

I used Claude Code as a coding assistant, to write and debug code and run the experiments I asked for. On NVIDIA's free tier, Nemotron 3 Ultra picks and judges, Mistral Large 4 gives the second opinion, and MiniMax M3.1 audits at the end. The calls that shaped the tool were mine: staying on free NVIDIA models, holding the doc until the quality numbers improved, doing the blind A/B myself (my three "no" votes became the craft checks), and asking for a prototype that looks and reads like a real Kuku tool.

Where they got it wrong: the model that picked the clips also graded them, and said my changed version was better when it was worse. Only judges from other model families caught it, so I no longer let a picker grade its own work. Smaller misses: the first models suggested had been retired (HTTP 410), and the "post first" rule once put a clip with a weak cliffhanger at #1 because it checked ad safety before quality.